

STRATOSERP

Master Feature Specification

An integrated, AI-powered campus ERP platform — Next.js · Node.js · MySQL · Gemini API

9	4	5	Next.js + MySQL
Core Features	Role Types	AI Integrations	Tech Stack

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CORE ARCHITECTURE & IDENTITY

StratosERP is built on a modern, scalable full-stack architecture designed for institutional reliability and developer agility. All modules operate under a unified identity protocol and a globally-configured semester context.

Layer	Technology
Frontend	Next.js · TypeScript · Tailwind CSS
Backend	Node.js · Express
Database	MySQL
Object Storage	MinIO
AI / ML	Gemini API

Configuration Rules

- **Global Semester Configuration:** Admin sets the active semester type (ODD/EVEN) and universal start/end dates governing all timetable and attendance logic.
- **Student Identity Protocol:** Students are strictly identified via institutional Email ID and a standardised UID string — Format: `[StartYear]-[ClassDivisionRollNo]-[EndYear]`.

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AI-POWERED INTELLIGENCE — GEMINI API INTEGRATION

Gemini AI is embedded throughout the platform to automate administrative communication, surface personalised insights, intelligently triage support tickets, and tag content — reducing manual overhead for all stakeholders.

Smart Notice Generation	Admins and HODs prompt the AI to draft formal, formatted institutional notices (e.g. low-attendance warnings) ready for instant PDF distribution.
Teacher Guardian Assistant	Faculty feed grades and attendance into the AI to auto-generate personalised 'Areas of Improvement' reports for parent-teacher meetings.
Notice Auto-Tagging	The AI scans uploaded broadcast notices and generates UI filter tags such as [URGENT] and [HACKATHON] for quick discovery.
Grievance Triage & Routing	The AI categorises the severity of student grievance tickets and routes them directly to the appropriate authority, eliminating unnecessary escalations.
Syllabus Pacing Analyzer (Optional)	AI compares a faculty member's logged lecture topics against the master syllabus PDF to calculate real-time semester pacing status.

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ROLE-BASED ACCESS CONTROL (RBAC) & CAPABILITIES

StratosERP enforces a four-tier RBAC hierarchy. A single faculty user can simultaneously hold multiple designations, each granting distinct data access and action rights.

System Administration (Admin)

- Bulk CSV ingestion: Students, Faculty, Master Timetable, Semester Planner.
- Define college-wide global thresholds (e.g. max AICTE points).
- Access macro analytics — performance and attendance for the entire college.
- Change the active semester/year, triggering batch progression and alumni automation.

Branch Leadership (HOD)

- Assign subjects and classes to faculty for the active semester.
- View aggregated department analytics: performance, attendance, placements, exams.

Faculty — Multi-Designation Support

- Class Incharge: analytics, all marks, AI reports, and parent/TG contact info for their class.
- Subject Incharge: input mid-term and end-semester marks; subject-level analytics.
- Teacher Guardian (TG): manage ~20 students, award AICTE points, direct parent communication.

End User (Student)

- Academic Dashboard: personal marks, attendance, fees, AICTE points, and carry-forward subjects.
- Schedule Management: full weekly timetable and daily lecture breakdown.
- Live Faculty Locator: real-time teacher location via timetable x system clock.
- Grievance Portal: submit structured complaints and track resolution status.

4 ACADEMIC & SCHEDULING ENGINE

The scheduling engine provides deterministic, automated timetable generation with intelligent leave management and detailed lecture execution logging.

01 Deterministic Timetable Generation

A backend script loops a 1-week master CSV across the entire semester, cross-referencing the Academic Calendar to automatically skip holidays.

02 Planned Leave Management

Faculty select future absent dates and manually assign a substitute teacher for each individual lecture slot in advance.

03

Emergency Leave Auto-Assignment

Impromptu absences trigger a constraint-satisfaction algorithm: scans the branch first, then the whole college, assigns the substitute, and fires a nodemailer alert.

04

Lecture Execution Logs

After each class, the Subject Incharge logs exact Syllabus Topics and Additional Topics taught — tracking real-world delivery against the planned syllabus.

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CAMPUS OPERATIONS & COMMUNICATION

A unified communication layer enables tiered notice broadcasting and secure study-material distribution, with MinIO providing scalable object storage.

Tiered Notice Board

Staff broadcast opportunities (hackathons, events) with MinIO attachments. Admins can target the whole INSTITUTE or select branches; HODs/Faculty are restricted to their assigned BRANCH.

Study Material Distribution

HODs and Faculty upload class notes and PDFs directly to MinIO. Access is routed dynamically — a teacher can grant access to a specific class or an entire year level.

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ADVANCED ADMINISTRATIVE MODULES**Dynamic Exam Seating & Invigilation Matrix**

- **Seating:** Admin inputs classroom capacities and the exam schedule. A constraint algorithm dynamically generates seating charts ensuring students from the same branch/year do not sit adjacently.
- **Invigilation:** The system auto-assigns available faculty to exam supervision duties, load-balancing assignments to ensure equitable workload distribution.

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PEER-TO-PEER DISPUTE & GRIEVANCE RESOLUTION

A formalised system for handling campus issues quietly and efficiently, ensuring every grievance reaches the right authority without cluttering intermediate inboxes.

Student Submission Flow

Step	Actor	Action
1	Student	Submits a structured grievance form (category, description, evidence).
2	Gemini AI	Analyses severity, classifies ticket type, and selects the correct authority.
3	System	Routes ticket directly to the identified authority with a notification.
4	Authority	Reviews, responds, and marks the ticket Resolved / Escalated.
5	Student	Receives resolution update on their dashboard in real time.

Routing Intelligence

Grievance Category	Auto-Routed To
Lab equipment failure	Subject Incharge of that lab
Academic marks dispute	Subject Incharge → HOD (if unresolved)
Infrastructure / civil issue	Admin
Interpersonal / peer conflict	Class Incharge + Teacher Guardian
Fee / financial concern	Admin

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STUDENT SUBJECT DASHBOARD — SUPPLI / KT LOGIC

The student dashboard surfaces all active subjects for the current semester and automatically carries forward any failed subjects (KT or Supplementary) from prior semesters, giving the student a complete, consolidated academic view.

Subject Visibility Rules

Rule	Condition	Result on Dashboard
Regular	Subject belongs to current semester	Shown as Active.
KT (Keep Term)	Student failed subject in a previous semester	Carry-forwarded and shown with a [KT] badge until cleared.
Supplementary	Student has a supplementary exam pending	Shown with a [SUPPLI] badge; marks updated after result de
Cleared	KT / Suppli subject subsequently passed	Badge removed; subject archived to transcript view.

Implementation Notes

- A student's subject list is computed at login by unioning current-semester subjects with any unresolved KT/Suppli records from prior semesters.
- Subject Incharges can input Suppli marks independently of the regular marks cycle; the dashboard updates automatically on save.
- The [KT] and [SUPPLI] badges are also visible to the student's Class Incharge, TG, and parents for proactive follow-up.
- Multiple backlogs across different semesters are all displayed simultaneously — no subject is hidden until cleared.

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ACADEMIC PROGRESSION & ALUMNI AUTOMATION

When the Admin advances the active semester, the system automatically re-computes every student's academic year and, for final-year graduates, transitions them to a dedicated Alumni section — all without manual data migration.

Semester-to-Year Mapping

Semesters Completed	Student Year Label	Section
1 – 2 (Sem 1 & 2)	1st Year	Active Students
3 – 4 (Sem 3 & 4)	2nd Year	Active Students
5 – 6 (Sem 5 & 6)	3rd Year	Active Students
7 – 8 (Sem 7 & 8)	4th Year	Active Students
8 completed (Graduated)	Alumni	Alumni Section

Progression Logic

Trigger	Admin changes the active semester — e.g. from Sem 2 (EVEN) to Sem 3 (ODD). This single action kicks off the batch progression pipeline.
Year Recalculation	Every active student's academic year label is recomputed based on their admission batch's semesters completed. A batch that just finished Sem 2 becomes 2nd Year when Sem 3 activates.
Alumni Transition	Students whose batch has completed Sem 8 are automatically moved to the Alumni section. Their academic records are archived; they lose access to timetables and attendance but retain read-only access to transcripts and results.
KT / Backlog Handling	Students with uncleared KTs are NOT auto-progressed to Alumni. They remain as active students with carry-forwarded backlog subjects until all are cleared.
Alumni Profile	Alumni records display graduation year, branch, final CGPA, and any placement data — accessible to Admin and HOD for institutional reporting.